

Advanced International Macroeconomics B (EI136)

Class 3

Global financial cycle

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Outline of today class

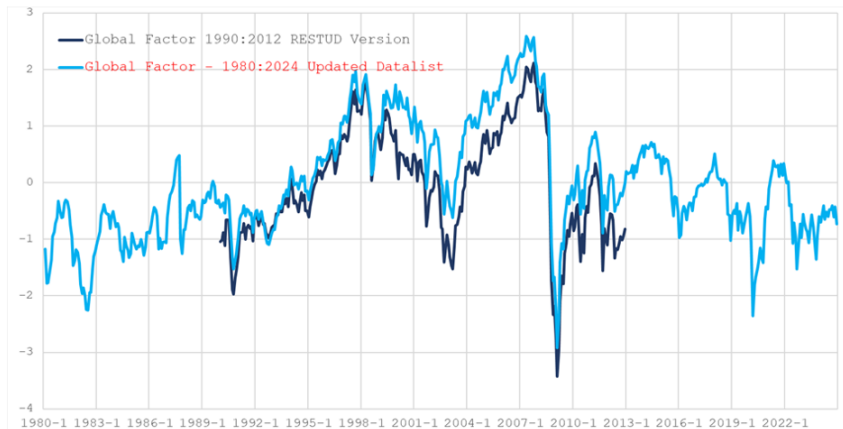
- Global financial cycle.
 - Pattern (Miranda-Agrippino and Rey 2021).
 - Impact on capital flows (Davis, Valente, and van Wincoop 2021, Scheubel, Stracca and Tille 2024).
 - Connection of exchange rate and risk (Georgiadis, Müller, and Schumann 2024).
- Financial constraints and the international transmission of shocks.
 - Pecuniary externality (Arce, Bengui and Bianchi 2023).
 - Endogenous frictions in foreign exchange market (UIP premium, Akinici and Qeralto 2024).
- Trilemma of policy in open economies.
 - Measures (Aizenman, Chinn and Ito 2022).
 - Empirical and theoretical assessments (Han and Wei 2017, Cesa-Bianchi, Ferrero, and Li 2025).
 - Limited use of exchange rate when there are not enough tools (Jeanne 2022).

GLOBAL FINANCIAL CYCLE

- Extract the common components of financial variables through a dynamic factor model (Miranda-Agrippino and Rey 2021).
- Price of **risky assets**: gives the Global Financial Cycle.
 - First component explains 24% of volatility.
 - Strong correlation with VIX and various measures of risk and risk aversion.
- VAR analysis of shock to US monetary policy.
 - A tightening lowers the first component of global asset prices and capital flows.
 - Effect similar before and after 2008.
- Policy of the ECB and PBoC also matters, but more through the Global trade cycles.

GFCy measure

- First component in risky (equity) asset prices.



Miranda-Agrippino, Silvia, and Hélène Rey (2021). "The Global Financial Cycle," NBER Working Papers 29327

Updated series: <https://silviamirandaagrippino.com/code-data>

<https://www.dropbox.com/scl/fi/aliquekgv9yzlfdmwyqcn/GFC-Factor-Updates-2024.xlsx?rlkey=4hrbknfue1q1l8y7hntffj008&st=o1scqels&dl=0>

- Are capital flows driven by global factors? Do the drivers of gross flows also drive net flows (not obvious they should)? Does this connect to the GFCy?
- Use a latent factor model do extract the common comovements across 58 countries (Davis, Valente, van Wincoop 2021).
 - Strong correlation of net and gross flows with world flows points to a common driver.
 - Loadings on factors vary across countries.
- Two factors account for 40-50% of flows. The first one reflects the price of risky assets (related to Miranda-Agrippino and Rey). The second one reflects commodity prices.
- The first factor matters for advanced and emerging (inflows), the second for emerging outflows (commodity exporters). The factors matter also for net flows.

- Factors reflects financial conditions and commodity prices.

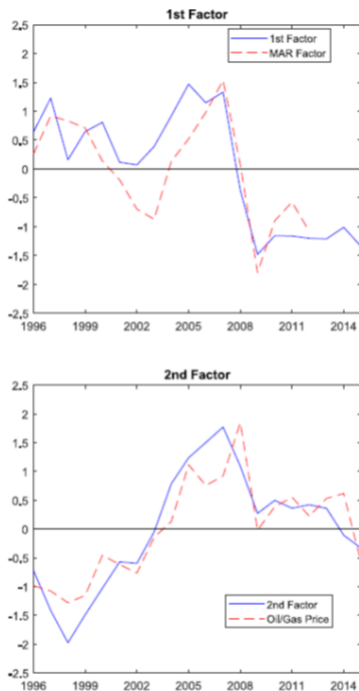


Fig. 3. First and Second Global factors.

Davis, Scott, Giorgio Valente, and Eric van Wincoop (2021). "Global Drivers of Gross and Net Capital Flows", *Journal of International Economics* 128.]

- First factor matters broadly for flows, second for EM outflows.

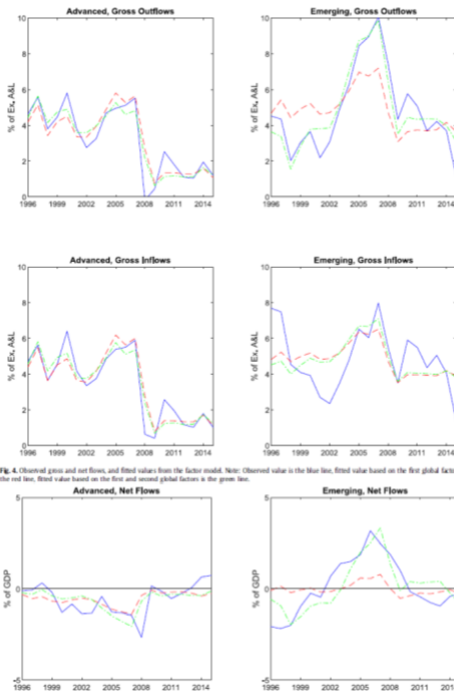


Fig. 4. Observed gross and net flows, and fitted values from the factor model. Note: Observed value is the blue line, fitted value based on the first global factor is the red line, fitted value based on the first and second global factors is the green line.

Fig. 4 (continued).

Davis, Scott, Giorgio Valente, and Eric van Wincoop (2021). "Global Drivers of Gross and Net Capital Flows", *Journal of International Economics* 128.

Heterogeneous sensitivity to factors

- The loadings are quite different across countries.
- They are asymmetric for outflows and inflows, hence also matter for net flows.
- Cross sectional regression of the loadings (first factor) on outflows, inflows, and net flows.
 - More sensitivity if integration (assets + liabilities) is high, especially in bonds and banks.
 - Less sensitivity if net creditor, especially in bonds and banks.
 - More outflows sensitivity if peg, and capital mobility.

Sensitivity depending on positions

- Integration (**asset + liabilities**) raises sensitivity, net creditor position (**asset - liabilities**) reduces it.

Table 5

Results from regressions of the GFC factor loadings on macroeconomic variables.

	(1)	(2)	(3)	(4)	(5)	(6)
	λ_1^{out}	λ_1^{out}	λ_1^{in}	λ_1^{in}	$\lambda_1^{out} - \lambda_1^{in}$	$\lambda_1^{out} - \lambda_1^{in}$
$A + L$	0.26*** (0.09)		0.18 (0.12)		0.08 (0.08)	
$A^{debt} + L^{debt}$		0.65*** (0.19)		0.64** (0.26)		0.01 (0.20)
$A^{FE} + L^{FE}$		0.46 (0.91)		-0.25 (1.28)		0.71 (0.97)
$A^{FDI} + L^{FDI}$		-0.28 (0.22)		-0.42 (0.31)		0.14 (0.24)
$A - L$	-0.96*** (0.34)		-2.04*** (0.45)		1.08*** (0.31)	
$A^{debt} - L^{debt}$		-0.92*** (0.33)		-1.85*** (0.46)		0.93*** (0.35)
$A^{FE} - L^{FE}$		1.42 (1.08)		0.46 (1.51)		0.96 (1.14)
$A^{FDI} - L^{FDI}$		-1.04 (1.09)		-0.98 (1.53)		-0.06 (1.16)
Inflation	0.05* (0.03)	0.04* (0.02)	0.00 (0.03)	-0.01 (0.03)	0.05** (0.02)	0.05** (0.03)
Peg	0.82** (0.39)	1.19*** (0.37)	0.57 (0.52)	0.89* (0.52)	0.25 (0.36)	0.30 (0.39)
CapControls	-1.30* (0.69)	-1.27** (0.61)	-0.39 (0.91)	-0.27 (0.86)	-0.90 (0.62)	-1.00 (0.65)

Davis, Scott, Giorgio Valente, and Eric van Wincoop (2021). "Global Drivers of Gross and Net Capital Flows", *Journal of International Economics* 128.

- Consider a broad set of GFCy measures, and assess the impact on (large) capital flows (Scheubel, Stracca and Tille 2024).
 - VIX index.
 - Global Stock Market Factor (similar to Miranda-Agrippino and Rey factor).
 - Excess Bond Premium.
 - Broad nominal exchange rate of the US dollar.
 - Broker-dealer leverage.
- Various measures comove, but with some heterogeneity.
- Tighter GFCy affects the probability of nonresident-driven capital flows.
 - Higher probability of a sudden stop.
 - Lower probability of a surge.
 - Little impact on resident-driven flows (flight and retrenchment).

Impact beyond capital flows

- Consider a broad range of financial variables (asset prices, credit volumes, international capital flows) in small advanced and emerging economies (Cerutti and Claessens 2024).
- For each variable in a country, regress it on:
 - Common factor of that variables across countries in its group (advanced or emerging).
 - GFCy variable (out of several possible ones): Vix, dollar broad exchange rate, Fed fund rate (nominal or real), Ted spread, yield curve slope, US growth, US money growth.
 - Look at the distribution of R2.
- Larger explanatory power for interest rates (75%), then for other asset prices (40-60%), followed by credit volumes (30%) and capital flows (25%).

RISK AND THE DOLLAR

Safe haven nature of the U.S.

- When risk goes up, investors move to US assets and appreciate the dollar. How does the exchange rate amplify the effect of risk shocks (Georgiadis, Müller, and Schumann 2024).
- Bayesian VAR with:
 - US and rest-of-world industrial production, and prices.
 - US and rest-of-world industrial policy interest rate.
 - Nominal effective dollar exchange rate.
 - Excess bond premium (risk price), bank credit.
- Identify by high frequency movement of gold prices around events pertaining to US or global risk.
 - Captures the price of risk.
- Shocks lead to a dollar appreciation, a global recession, credit contraction, and rise in spread.

Effect of risk shock

- **Appreciation** of the dollar, **recession**, and higher **spread**. Also contraction in bank credit.

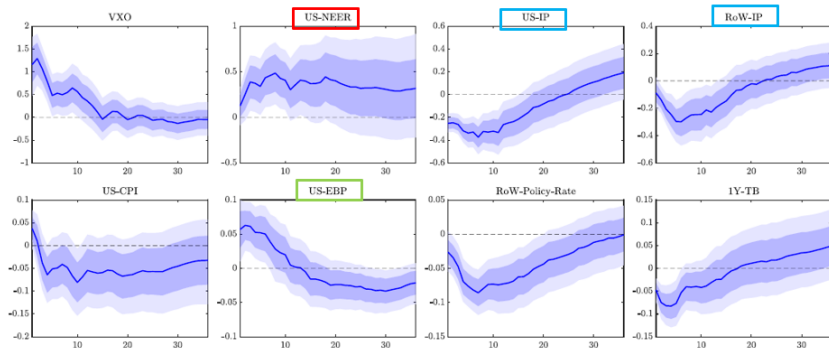


Fig. 2. Impulse responses to a global risk shock.

Note: Horizontal axis measures time in months, vertical axis deviation from pre-shock level; size of shock is one standard deviation; blue solid line represents point-wise posterior mean, light-shaded areas 90% and dark-shaded areas 68% equal-tailed, point-wise credible sets. VXO measured in levels, the dollar NEER, US and RoW industrial production, US consumer prices in logs, and the excess bond premium, the RoW policy as well as the US 1-year Treasury Bill rates in percent.

- Effect of the risk shocks when shutting down the exchange rate.
 - Additional shocks that stabilize the exchange rate.
 - Stabilization of the exchange rate by the Federal Reserve.
 - Removal of financial amplification channel in a US-RoW model.
- Without the exchange rate movement, the recession is reduced (by 30-50%).
 - Not through the trade expenditure channel (the trade balance does not change much).
 - Through the financial channel with smaller risk premia and a more moderate credit contraction.

- Exchange rate is steady, recession is smaller (dots in third column: model calibrated with and without financial amplification).

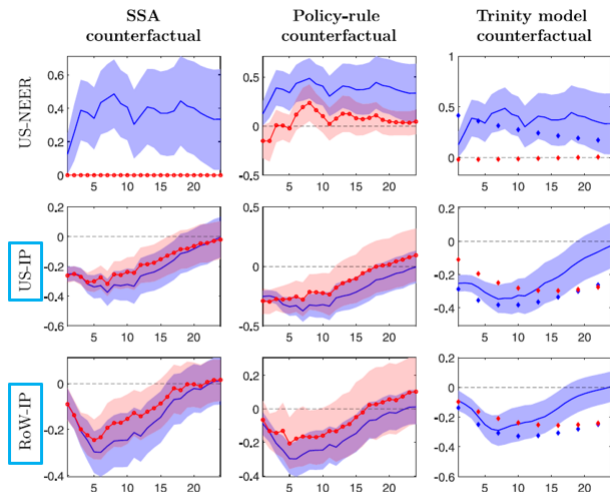


Fig. 4. Baseline and counterfactual responses to a global risk shock.

Georgiadis, Georgios, Gernot Müller, and Ben Schumann (2024). "Global risk and the dollar" *Journal of Monetary Economics* 144

- Without the dollar appreciation, spreads increase by less (actually fall) and credit contracts much less.

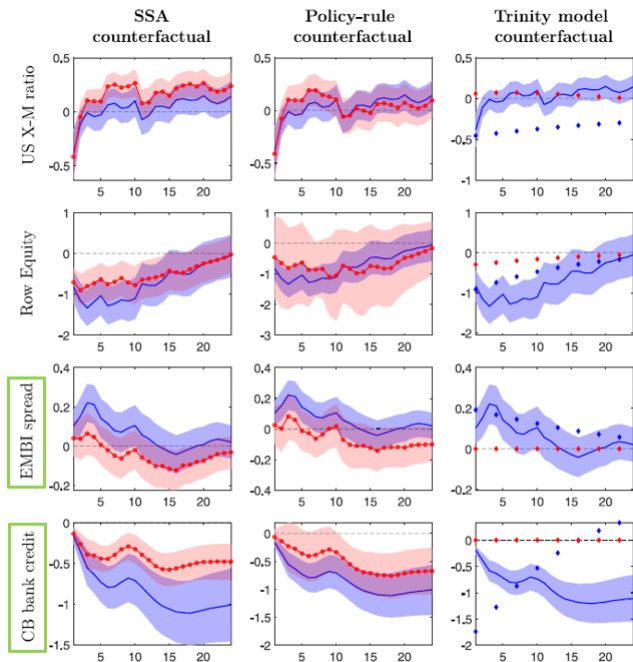


Fig. 5. Baseline and counterfactual responses of trade and financial variables to a global risk shock.

Georgiadis, Georgios, Gernot Müller, and Ben Schumann (2024). "Global risk and the dollar" *Journal of Monetary Economics* 144

PECUNIARY EXTERNALITY

(Ultimately) costly credit booms

- A credit boom, financed by foreign funds, raises vulnerability to fall in collateral value.
 - Without frictions this is not an issue. With a borrowing constraint, it becomes one.
- In an model with traded and non-traded goods, agents want to borrow, but debt is constrained in proportion of the traded value of output.
 - Real exchange rate acts as a collateral price. A real depreciation reduced the value of non-traded endowment (expressed in traded good).
- Households treat the real exchange rate as exogenous. The planner knows that it is endogenous.
 - Lower borrowing by the planner to limit the subsequent exchange rate depreciation (constraint tightening) in case of bad news.
 - Can be implemented with a tax on borrowing.

Ex ante and ex post policies

- Simple production model where the planner can tax borrowing ex-ante, and reallocate labor between traded and non-traded production ex post (Arce, Bengui and Bianchi 2023).
- Small open economy accessing a bond with return in terms of traded goods. Borrowing constraint (not always binding) in proportion of national income (wage and profits in the two sectors):

$$\frac{b_{t+1}}{R} \leq \kappa \left(w_t \bar{h} + \pi_t^T + \pi_t^N \right)$$

- The non-traded profits are affected by the price of the non-traded good (real exchange rate).
- With a bad shock the constraint binds. Consumption of traded must fall, i.e. the non-traded good has to become cheaper, reducing the collateral value.

Decentralized vs. centralized allocation

- Agents take the real exchange rate as exogenous.
- The central planner does not, and internalize the effect of a binding constraint on the collateral value.
 - Limit initial borrowing by taxing debt ex-ante, when the constraint is not binding.
- What if the planner can also reallocate labor ex-post? Shift it towards the traded sector, through a labor tax in the non-traded sector.
- Numerical allocation: the ex-ante tax on debt is always warranted.
 - Even when ex-post policy is available, leading to a higher debt (the economy can handle more debt when in a crisis).
 - The agents always take the collateral price (real exchange rate) as given.

Rules with ex ante policy

- Binding constraint is the right part of each panel.
- Lower debt under **ex ante policy** than under **decentralized** allocation, but coincide once crisis hit.

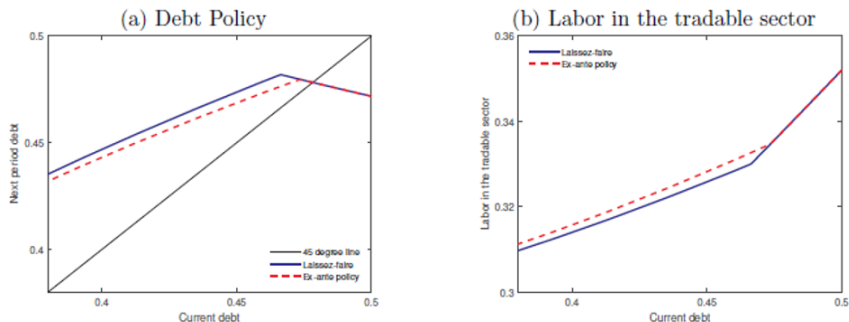


Figure 1: Policy functions: ex ante intervention

Arce, Fernando, Julien Bengui, and Javier Bianchi (2023). "Overborrowing, Underborrowing, and Macprudential Policy," CEPR Discussion Paper 18273.

Expanding to ex post policy

- Allocation under **policy** differs from **decentralized** even when the constraint binds.
- Policy shifts labor towards traded sector ex-ante, and can afford more debt.

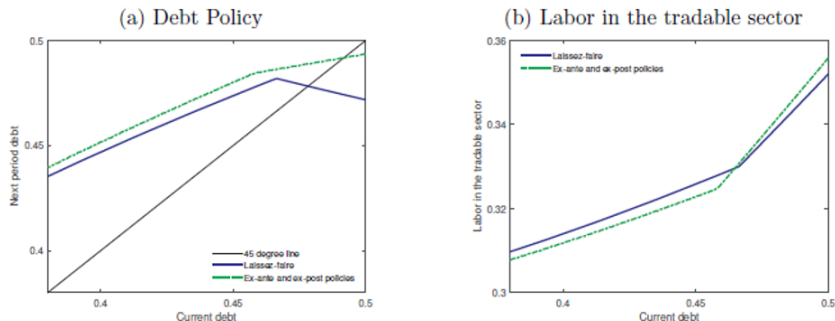


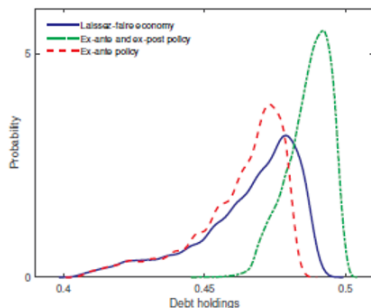
Figure 2: Policy functions: ex ante and ex post intervention

Arce, Fernando, Julien Bengui, and Javier Bianchi (2023). "Overborrowing, Underborrowing, and Macropudential Policy," CEPR Discussion Paper 18273.

Taxation of debt

- Even though debt is higher when both policies are available, it is still taxed.
- Without that, agents would undermine the ex post policy by borrowing even more.

(a) Density distributions of debt



(b) Taxes on debt

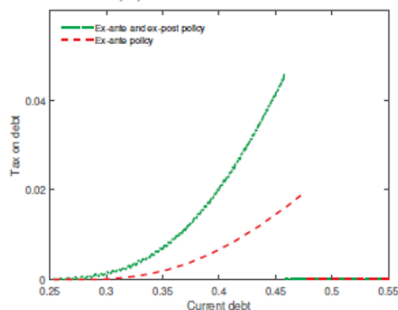


Figure 3: Policy functions for debt and labor in the tradable sector

Arce, Fernando, Julien Bengui, and Javier Bianchi (2023). "Overborrowing, Underborrowing, and Macroprudential Policy," CEPR Discussion Paper 18273.

Endogenous interest parity deviation

- Small open economy, with lending through banks. Invest in capital, funded by equity and domestic and foreign borrowing (Akinci and Queralto 2024).
- Leverage constraint, to prevent bankers from running with the money.
 - Limits borrowing, hence return on capital $>$ interest rate on bonds.
 - Running easier to do when funding is from abroad, hence tighter constraint. Factor $\gamma > 0$: extra tightening of leverage constraint with foreign borrowing (instead of domestic).
- Return spread: μ_t between capital return and domestic interest rate, ϱ_t between capital return and foreign interest rate (adjusted for exchange rate dynamic), and μ_t^* between domestic and foreign rates (UIP deviation).
 - Higher spread vis-à-vis foreign rate, because of higher diversion risk:
 $\varrho_t = (1 + \gamma) \mu_t$
 - UIP deviation: $\mu_t^* = \varrho_t - \mu_t = \gamma \mu_t$

Dynamics in a simple model

- Domestic spread μ_t higher when banks have lower equity, or a higher share of foreign funding.
- Capital (collateral) price inversely related to spread μ_t .
- Current exchange rate more depreciated when UIP spread $\gamma\mu_t$ is higher.
- Effect of a drop in banker's net worth.
 - Higher premium μ_t and lower capital price.
 - Higher UIP premium if $\gamma > 0$.
- Amplification if $\gamma > 0$ and there is foreign funding: low net worth shock $\rightarrow$ higher spread $\mu_t \rightarrow$ higher UIP deviation $\gamma\mu_t \rightarrow$ extra depreciation of the exchange rate $\rightarrow$ higher value of foreign currency debt (in domestic currency) $\rightarrow$ further reduction in net worth.

Net worth shock

- Effect on UIP premium and exchange rate.

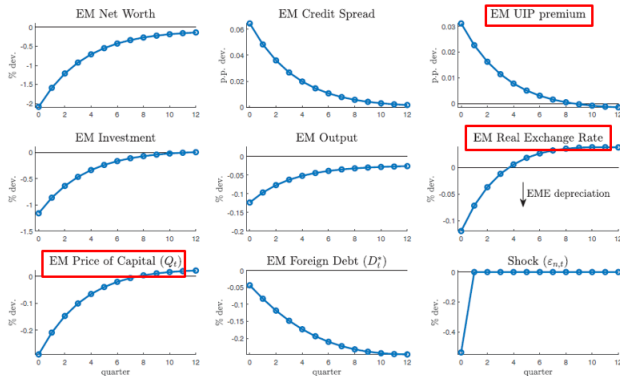


Figure 2
Shock to bankers' net worth

Effects of a negative shock to bankers' net worth.

Akinci, Ozge, and Albert Queralto (2024). "Exchange Rate Dynamics and Monetary Spillovers with Imperfect Financial Markets," *The Review of Financial Studies* 37(2), pp. 309–355.

- Standard calibrated DSGE model, to study the impact of a higher US interest rate.
- Without financial friction: small decrease in output of small open economy.
 - Positive expenditure switching effect from exchange rate depreciation offsets expenditure reducing effect.
- With financial friction, stronger transmission of US shock.
 - Dollar debt leads to UIP deviation, and amplifies effect on return spread, investment and output.
- Transmission stronger under dollar invoicing, as the exchange rate depreciation does not leads to positive expenditure switching.
- Despite amplification effect through UIP, pegging the exchange rate is a bad idea.
 - Need to move the interest rate a lot, as its effect on the exchange rate is moderate.
 - Destabilizes real variables.

Impact of US policy

- Amplification from dollar debt and financial constraint.

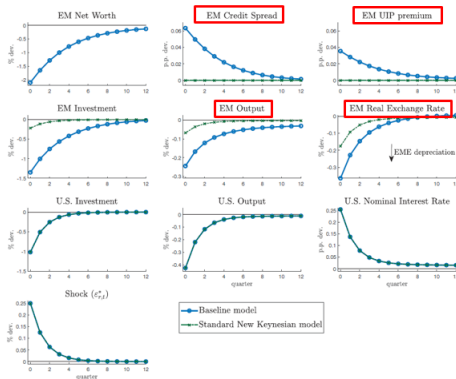


Figure 1
U.S. monetary policy shock

Effects of a rise in the U.S. monetary shock $\epsilon_{T,t}^*$ in (67) of 0.25%. The blue line with circles represents the effect in our baseline model, and the green line with crosses represents the effects in the standard New Keynesian model.

Akinci, Ozge, and Albert Queralto (2024). "Exchange Rate Dynamics and Monetary Spillovers with Imperfect Financial Markets," *The Review of Financial Studies* 37(2), pp. 309–355.

TRILEMMA OR DILEMMA?

Trade-offs in international policy

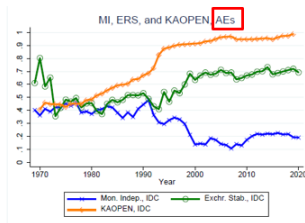
- Standard Trilemma of international macroeconomics: one can have only two out of three goals: a) independent monetary policy, b) stable exchange rate, c) capital mobility.
- Indices measuring the policy choice (Aizenman, Chinn and Ito 2022).
 - Exchange rate stability (*ERS*), inverse of variance of FX vis-à-vis a reference currency.
 - Monetary independence (*MI*), sensitivity of policy rate to rate in a reference country (US or Euro).
 - Capital account openness (*KAOPEN*), “Chinn-Ito” index of capital mobility.
- Evolution through time is contrasted across country groups.
 - Advanced: more KO, more ERS and less MI (driven by Euro).
 - Emerging: more KO, less ERS.
 - Developing: no major changes.

Indices of trilemma corners

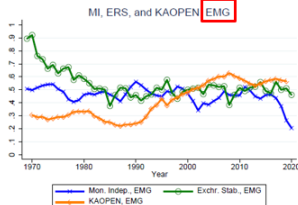
- Contrasted evolution across country groups.

Figure 2: Development of the Trilemma Configurations Over Time

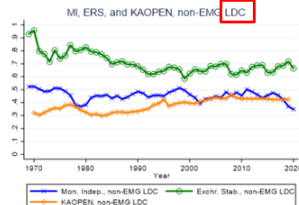
(a) Advanced Economies



(b) Emerging market economies



(c) Non-Emerging Market Developing Countries



Aizenman, Joshua, Menzie Chinn, and Hiro Ito (2022). "The Impacts of Crises on the Trilemma Configurations", NBER working paper 30406

Exchange rate regimes and policy autonomy

- Group countries according to FX regime (IMF classification) and capital mobility (Chinn-Ito index).
- Regression of changes in domestic policy interest rate on changes in anchor (US, euro area) policy interest rate. Autonomy is a small, or zero, coefficient.
 - Floating exchange rate raises autonomy (smaller coefficient on anchor), with gradual effect as a partial float already helps.
 - Capital controls raise autonomy, but need to be comprehensive (no effect of partial controls).
- Need to control for other (possibly common) drivers of the interest rates (Han and Wei 2017). Change in a country's rate reflects:
 - (Estimated) Taylor rule, because of changes in growth and inflation prospects (from WEO).
 - US rate, measuring the extent of policy autonomy (inverse).
 - Risk (VIX).
- The coefficient on the US rate should differ depending on the degree of capital mobility (Chinn-Ito index) and the exchange rate regime.

- Policy autonomy with capital controls, the interest rate does not follow the US rate.
- Without controls, a flexible exchange rate offers *some* autonomy.
 - Connection stronger after 1998, reflecting higher financial integration. Observed for short maturity and long maturity interest rates.
- Allow for a different connection to US interest rate depending on whether it increases or decreases.
 - No connection to rising US rates under a flexible exchange rate regime (let the domestic currency depreciate).
 - Connection to falling US rates (counter the appreciation of the domestic currency).
- Overall a “2.5-lemma” instead of a trilemma vs. dilemma.
 - Capital controls deliver autonomy. Flexible exchange rate delivers autonomy only when the US interest rate increases, but not when it decreases (fear of local currency appreciation).

- Autonomy (no sensitivity to US rate) when there are **capital controls**, regardless of FX regime.
- Without **capital controls**, more sensitivity to US rate under a fixed exchange rate than under a flexible one.

Table 3

Baseline model for both short-term policy rates and long-term government bond yields.

		Short-term 1990–2009 (1)	Short-term 1990–1998 ^a (2)	Short-term 1999–2009 ^b (3)	Long-term 1999–2009 (4)
$r_{i,t}^p - 1$	λ	−0.048 ^e (0.008)	−0.007 (0.015)	−0.11 ^e (0.01)	−0.071 ^e (0.013)
$\Delta GDP\ growth_{i,t}$	ϕ_1	0.096 (0.06)	0.237 (0.144)	0.044 (0.053)	0.027 (0.025)
$\Delta Inflation_{i,t}$	ϕ_2	0.329 ^e (0.048)	0.134 (0.096)	0.413 ^e (0.049)	0.172 ^e (0.039)
$D_{fixed,NC}\Delta r_{i,t}^{US}$	β_1	0.649 ^e (0.39)	0.402 (2.09)	0.648 ^e (0.297)	0.665 ^e (0.259)
$D_{fixed,C}\Delta r_{i,t}^{US}$	β_2	0.034 (0.325)	1.998 (1.286)	−0.047 (0.262)	0.118 (0.302)
$D_{flex,NC}\Delta r_{i,t}^{US}$	β_3	0.450 ^e (0.176)	0.492 (0.438)	0.493 ^e (0.153)	0.473 ^e (0.111)
$D_{flex,C}\Delta r_{i,t}^{US}$	β_4	0.029 (0.127)	0.008 (0.334)	0.059 (0.109)	0.251 ^e (0.099)
ΔVIX_t	δ	0.23 (0.199)	0.086 (0.584)	0.172 (0.168)	0.177 ^e (0.089)
$t\text{-Test: } \beta_1 > \beta_3$		−0.473	0.042	−0.472	−0.682
$t\text{-Test: } \beta_4 < \beta_3$		1.998 ^e	−0.892	−2.416 ^e	−1.514
Adjusted R-squared		0.09	0.00 ^d	0.30	0.22
No. of Obs.		827	295	531	356

^a During 1990–1998, with available policy rate data, there is only one economy observed for the regime of fixed exchange rate and no capital controls (HKG).

^b Argentina 2001/April is an outlier and removed from the regression sample.

^c To test $\beta_1 > \beta_3$, we re-write the regression as $\Delta r_{i,t}^p = c + \lambda r_{i,t-1}^p + \phi_1 * \Delta GDP\ growth_{i,t} + \phi_2 * \Delta Inflation_{i,t} + \beta_1 (D_{fixed,NC}\Delta r_{i,t}^{US} + D_{flex,NC}\Delta r_{i,t}^{US}) + \beta_2 D_{fixed,C}\Delta r_{i,t}^{US} + (\beta_3 - \beta_1) D_{flex,NC}\Delta r_{i,t}^{US} + \beta_4 D_{flex,C}\Delta r_{i,t}^{US} + \delta \Delta VIX_t + e_{i,t}$. In this form, the condition that $\beta_3 < \beta_1$ is the same as the coefficient, $(\beta_3 - \beta_1) < 0$.

^d The unadjusted R-squared for Column 2 is 0.027.

^e Significant at the 10%.

- Without capital controls, sensitivity to US rate under a **fixed exchange rate**.
- With a flexible rate, no sensitivity to a US rate **increase** (appreciates the dollar), sensitivity to US rate **decrease** (depreciates the dollar).

Table 4

Asymmetric responses under flexible exchange rate regimes plus an alternative Taylor rule specification.

		(1) Asymmetric responses	(2) Adding exchange rate	(3) Only countries with policy rates ^a	(4) Long-term bond yield as dependent variable
$r_{i,t}^p - 1$	λ	-0.114* (0.03)	-0.117* (0.03)	-0.116* (0.03)	-0.069* (0.04)
$\Delta GDP\ growth_{i,t}$	ϕ_1	0.04 (0.07)	0.064 (0.08)	0.073 (0.08)	0.035 (0.03)
$\Delta Inflation_{i,t}$	ϕ_2	0.413* (0.09)	2.442* (1.36)	2.56* (1.33)	0.861 (0.57)
$\Delta Real\ Exchange\ Rate_{i,t}$	ϕ_3	- (0.17)	2.03 (1.34)	2.145 (1.31)	0.693 (0.57)
$D_{flex,NC} \Delta r_{i,t}^{IS}$	β_1	0.656* (0.17)	0.617* (0.17)	0.595* (0.17)	0.663* (0.19)
$D_{flex,C} \Delta r_{i,t}^{IS}$	β_2	-0.048 (0.15)	-0.039 (0.15)	-0.095 (0.14)	0.161 (0.14)
$D_{flex,NC} Pos\ \Delta r_{i,t}^{IS}$	$\beta_{3,pos}$	0.152 (0.23)	0.114 (0.24)	0.127 (0.25)	0.329* (0.12)
$D_{flex,NC} Neg\ \Delta r_{i,t}^{IS}$	$\beta_{3,neg}$	0.630* (0.160)	0.616* (0.170)	0.592* (0.17)	0.526* (0.12)
$D_{flex,C} Pos\ \Delta r_{i,t}^{IS}$	$\beta_{4,pos}$	0.215 (0.25)	0.192 (0.24)	0.169 (0.24)	0.001 (0.24)
$D_{flex,C} Neg\ \Delta r_{i,t}^{IS}$	$\beta_{4,neg}$	0.013 (0.16)	0.025 (0.16)	0.037 (0.16)	0.367* (0.22)
ΔVIX_t	δ	0.172 (0.15)	0.047 (0.18)	-0.032 (0.19)	0.116 (0.10)
Adjusted R-squared		0.304	0.307	0.303	0.223
No. of Obs.		531	531	514	356

Note: Robust standard errors clustered at the level of country-period are reported for all columns.

Bold and an asterisk means significant at the 10% level.

^a Argentina is excluded as it is an obvious outlier.

Han, Xuehui, and Shang-Jin Wei (2017) "International transmissions of monetary shocks: Between a trilemma and a dilemma" *Journal of International Economics* 110, pp. 205-219.

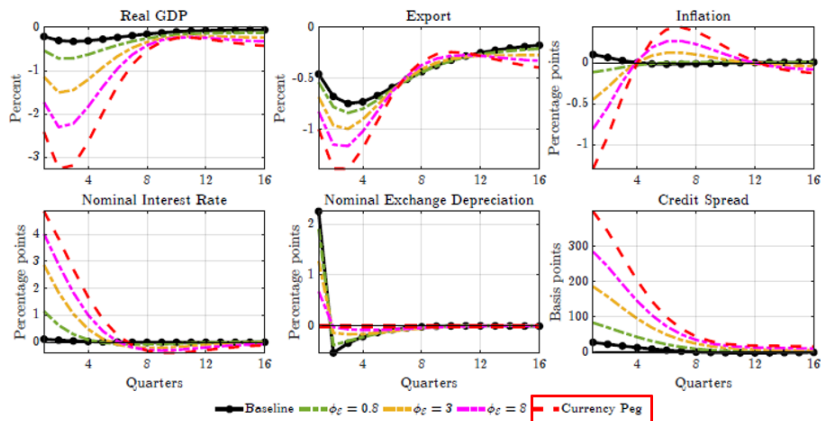
Exchange rate regime in a model

- Two country model with a financial accelerator channel (dollar debt) and the expenditure switching channel (Cesa-Bianchi, Ferrero, and Li 2025).
- Empirical evidence with multi-country VAR.
 - Tightening of US monetary policy is contractionary, in U.S. and rest of the world.
 - Contraction more limited for countries with flexible exchange rates.
- Two country model with domestic and financial frictions (similar to Akinci and Queralto 2024).
- Tightening of foreign (dominant currency) monetary policy raises spreads and lowers GDP.
 - A peg does not insulate, but moderates the impact.
 - Countercyclical taxes on domestic credit and / or foreign currency also limits volatility, especially under a float.

Impact of foreign policy tightening

- Recession and increase in spreads. Much less so (but not zero) under a floating exchange rate.

Figure 7: Comparing different degrees of exchange rate flexibility.



Cesa-Bianchi, Ambrogio, Andrea Ferrero, and Shangshang Li (2025). "Capital Flows and Exchange Rates: A Quantitative Assessment of the Dilemma Hypothesis", draft

Using other tools

- Macroprudential policy (tax on credit) and capital controls (tax on foreign borrowing) help.

Table 3: Exchange rate regimes, policy rules and macroeconomic volatility.

Regime (additional instrument)	Policy rules coefficients		Standard deviations	
	ϕ_{ε}	ϕ_k or ϕ_b	Real GDP	Inflation
Fully flexible exchange rate	0	0	1.22	0.07
Baseline	0.062	0	1.58	0.05
Baseline + tax on domestic credit (τ_k^{flex})	0.062	-0.05	0.95	0.07
Baseline + tax on foreign borrowing (τ_b^{flex})	0.062	-0.15	0.60	0.32
Peg	∞	0	109.66	8.79
Peg + tax on domestic credit (τ_k^{peg})	∞	-0.05	21.88	5.66
Peg + tax on foreign borrowing (τ_b^{peg})	∞	-0.35	22.46	4.77

Cesa-Bianchi, Ambrogio, Andrea Ferrero, and Shangshang Li (2025). "Capital Flows and Exchange Rates: A Quantitative Assessment of the Dilemma Hypothesis", draft

POLICY WITH UIP SHOCKS

Policy when tools are limited

- Why don't policy makers (especially in EM) make full use of exchange rate flexibility (Jeanne 2022)?
- Exchange rate movements reflect deviations from UIP:

$$(1 + i_t) \frac{E_t}{E_{t+1}} = (1 + i_t^*) (1 + \rho_t)$$

- Lower ρ_t implies a shift of investors towards local currency bonds (lower i_t).
- Simple small open economy model.
 - Endogenous production of traded and non-traded goods.
 - Wage stickiness in short run (1 period), then full flexibility.
 - Short run shock to UIP premium $\rho - \bar{\rho}$.
 - Inflows in local currency bonds (holdings B), outflows in dollar bonds (holdings B^*).

- Lower premium ($\rho - \bar{\rho} < 0$) leads to trade deficit ($x < 0$) and higher gross inflows and outflows:

$$x = b^* - b = \frac{1}{\sigma_T} (\rho - \bar{\rho})$$

$$b^* = -\frac{\phi}{\sigma_T} (\rho - \bar{\rho})$$

$$b = -\frac{1 + \phi}{\sigma_T} (\rho - \bar{\rho})$$

- Lower cost of borrowing, so increase \$ investment and borrow through a trade deficit.

Policy response

- Trade-off between output gap in traded sector (reflects exchange rate e) and non-traded sector (reflects interest rate i):

$$y_T = \frac{e}{\gamma} \quad ; \quad y_N = -\frac{i - \bar{i}}{\sigma_N}$$

- UIP ($i - \bar{i} + e = \rho - \bar{\rho}$) implies that we cannot set $i - \bar{i} = e = 0$.
- Optimal (limited) policy: some appreciation of the real exchange rate, some reduction in interest rate (less than foreign rate: $i - \bar{i} < \rho - \bar{\rho}$).
- Planner's optimum: lower trade deficit x and smaller capital inflows b , no change to outflows b^* .
- Can be implemented through a tax on holding domestic bonds for all investors. They get $(1 + i_t) / (1 + \tau) < 1 + \bar{i}$.
 - Higher tax when UIP shock is low:

$$\tau = -(1 - \chi)(\rho - \bar{\rho})$$

- Tax inflows and outflows. Taxing only inflows is very suboptimal.
- Using tax and exchange rate flexibility leads to better welfare.

Welfare impact

- Tax on investment has a strong effect than FX flexibility.

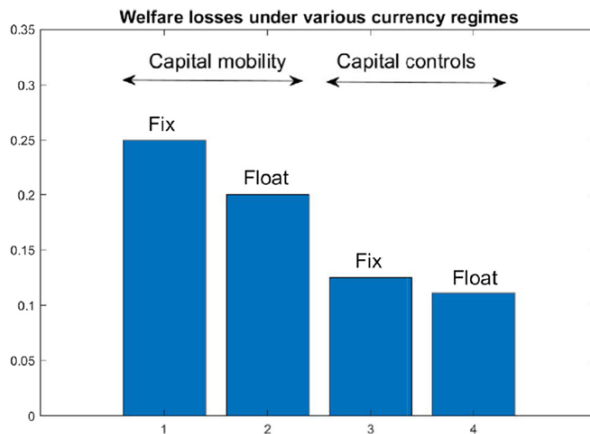


Fig. 2. Welfare losses under different policy regimes.

Jeanne, Olivier (2022). "Rounding the corners of the trilemma: A simple framework", *Journal of International Money and Finance* 122.